

Co-firing copper in a cordierite-type glass-ceramic printed by multi-material vat photopolymerisation: a model-based process design

[First Author]^{a,*}, [Second Author]^a

^a[Department, Institution], [City], [Country]

Abstract

Multi-material vat photopolymerisation can print a glass-ceramic body and its copper conductors as one green part, but whether such a part can be fired is not known. Carbon must leave the glass before its pores close, copper must stay metallic, and the two materials must shrink compatibly, all in one temperature range. A verified model couples pyrolysis and steam gasification of char, viscous sintering and crystallisation of the glass, diffusional sintering of copper, and co-firing mechanics, analytically and in three dimensions. Inputs come from the literature, and the glass was calibrated against the published IBM glass-ceramic/copper process. Of the two atmospheres compared, only steam with hydrogen held at a fixed fraction of the Cu/Cu₂O equilibrium, as in the IBM process, removes the char before the glass seals. Fine copper then densifies during the burnout, well before the glass, and the free-strain mismatch reaches 15.6%. A paste and programme optimised together use an assumed retarding effect of char on copper to release both materials at once, cutting the mismatch to 3.7%, but this design is fragile. A design that tolerates a factor of two in gasification rate and 15 K in the glass transition holds the mismatch at 9–11% and the copper at 89% IACS in a 16 h cycle. In three dimensions the two designs reduce the dishing of a printed test vehicle from 622 µm to 150–317 µm, and plastic flow of the copper keeps the cooling bow small. Gasification kinetics, glass viscosity, copper sintering viscosity and char yield should be measured first.

Keywords: Co-firing, Glass-ceramic, Copper metallisation, Vat photopolymerisation, Binder removal, Sintering mismatch

1. Introduction

Packages for millimetre-wave front ends, antenna-in-package modules and wide-bandgap power devices now have to route conductors in three dimensions through a low-loss dielectric, and to do it close to the die [1, 2, 3, 4]. Co-fired ceramics answer part of this demand. Low-temperature co-fired ceramic (LTCC) and glass-ceramic substrates combine low dielectric loss, hermeticity and a thermal expansion close to that of silicon [5, 6]. They are, however, built from stacked tapes: conductors are printed in planes, interconnections run vertically through punched vias, and every change of geometry requires new tooling [7, 5]. The Heterogeneous Integration Roadmap devotes a chapter to additive manufacturing for packaging, including lithography-based manufacturing of ceramic parts [8].

The conductor of choice in fired packages has long been silver, because it can be co-fired in air. Silver migrates under bias and humidity and diffuses into the glass when fired in oxidising conditions [9, 10, 11]. Copper conducts almost as well, costs less and is far less prone to migration, but it must be fired in an atmosphere that keeps it metallic. In such an atmosphere the organic binder no longer burns off cleanly and part of it is left as carbonaceous residue [12]; carbon sealed in a glass that has closed its pores can bloat or even burst the part [13]. IBM solved this for its glass-ceramic/copper multilayer substrates

with a steam-based co-sintering process that gave a dense glass-ceramic with high-conductivity copper [14, 6]. That solution was developed for laminated tapes.

Vat photopolymerisation of ceramic slurries, in particular lithography-based ceramic manufacturing (LCM), produces dense ceramics with features of a few tens of micrometres [15, 16, 17]. It has already been used for monolithic microwave filters and antenna substrates [18, 19, 20], and photocurable LTCC suspensions have been formulated for it [21]. In these parts the metallisation is still added afterwards, by plating or by printing on the fired ceramic [18, 22], or the conductors are jetted into ceramic layers by a separate route [23, 24, 25, 26]. Multi-material vat photopolymerisation removes that restriction. Printers such as the CeraFab Multi 2M30 switch between slurries within a layer [27, 28, 29]. Co-sintered ceramic–ceramic parts have been demonstrated [30], and zirconia and titania green bodies printed by vat photopolymerisation have been assembled and co-sintered in a reducing atmosphere, which made the titania conductive [31]. Copper slurries for vat photopolymerisation have been developed separately [32, 33, 34, 35]. In principle, then, a glass-ceramic body and its copper wiring can be printed as one green part. Copper and LTCC suspensions have already been inkjet-printed into a three-dimensional part and co-fired [23]. For vat photopolymerisation, whose slurries carry far more binder, we found no study of how such a part should be fired, and no model of co-firing that includes binder removal.

Four requirements make that firing difficult, and they are coupled. First, the char from both slurries must be removed before

*Corresponding author.

Email address: [email address] ([First Author])

the glass seals its pores, in an atmosphere that does not oxidise the copper. Second, the two materials must shrink at compatible times and by compatible amounts; otherwise the part cambers, delaminates or cracks around its conductors [36, 37, 38]. Third, the glass must densify before it crystallises, because crystals stop viscous flow. Fourth, a thermal-expansion mismatch of about 14 ppm K^{-1} between copper and a cordierite-type glass-ceramic must be accommodated on cooling. In tape-based LTCC the second requirement is usually met by adjusting the metal paste, for example with ceramic additions [39], or the heating schedule, guided by master sintering curves [40]. Continuum models of co-sintering reproduce camber and mismatch stresses in ceramic and metal–ceramic bilayers [41, 42, 43, 44, 45, 46], but none includes the chemistry of binder removal, and none treats copper with a crystallising glass. The first and third requirements are therefore usually handled separately from the second, although in practice they compete for the same temperature range.

This paper sets out a model-based design of that firing for a copper conductor printed inside a cordierite-type glass-ceramic body. We extend a slab model of binder removal and sintering, which we developed for single-material printed copper (Appendix A), to a crystallising glass-ceramic. The two materials are coupled through the viscous analogue of laminate theory and through a three-dimensional finite-element model of a voxelised printed part. The model is verified against closed-form solutions and parameterised from published data, with every input tagged by its provenance. We use it for four purposes. We (i) map the atmosphere and temperature window in which carbon can be removed from the glass before its pores close while copper stays metallic; (ii) show that fine copper and a fragile glass densify in different temperature ranges and with different curve shapes, and quantify the consequences for camber and conductor damage; (iii) design the copper paste and the firing programme together to bring the two into step, both at nominal inputs and robustly against their uncertainty, and follow the designs through a printed test vehicle in three dimensions; and (iv) identify the few material parameters that decide the outcome, and should therefore be measured first.

2. Model formulation

2.1. Structure of the model

The co-fired part is represented at two levels (Fig. 1b). At the first level, each material is followed through the firing programme by a one-dimensional slab model that resolves heat transfer, binder decomposition, char formation and removal, pore closure and densification through the thickness of its critical section. The glass-ceramic body and the copper conductor are simulated separately but share the furnace set point and the inlet gas, so both see the same thermal and chemical history. At the second level, the two densification histories are coupled mechanically: analytically, through the viscous analogue of laminate theory, and numerically, through a three-dimensional finite-element model in which both materials occupy the voxels of the printed part. Both levels use the same constitutive law for the porous body, so a free sample in the finite-element model reproduces the slab model exactly.

The copper slab model was developed and verified for single-material copper parts printed from a lithography-based slurry; its chemistry and its calibration against published copper data are summarised in Appendix A. The additions made here are the crystallising glass-ceramic, the copper paste with an inert filler, the steam burnout atmosphere, and the co-firing mechanics.

2.2. Binder removal and residual carbon

The photocured binder is lumped into three pseudo-components i (a non-reactive solvent, a crosslinked network and a char-forming backbone), each decomposing by first-order kinetics,

$$\frac{dB_i}{dt} = -k_i(T) B_i, \quad k_i(T) = \frac{E_i \beta_{\text{ref}}}{RT_{p,i}^2} \exp\left[-\frac{E_i}{R} \left(\frac{1}{T} - \frac{1}{T_{p,i}}\right)\right], \quad (1)$$

where $T_{p,i}$ is the temperature of the corresponding peak in a thermogravimetric trace at $\beta_{\text{ref}} = 10 \text{ K min}^{-1}$. Writing the rate constant in terms of the peak temperature keeps the two fitted quantities nearly independent and lets them be read directly from a single thermogravimetric curve. A fraction χ of the network and backbone mass remains as char. In a steam-bearing atmosphere the char is gasified by $\text{C} + \text{H}_2\text{O} \rightarrow \text{CO} + \text{H}_2$ at the rate

$$r_g = a(\varepsilon_o)(1 - f_{cl})k_g(T) \frac{c}{M_C} \left(\frac{x_{\text{H}_2\text{O}}}{x_{\text{ref}}}\right)^{1/2} \times \frac{1 + K_H x_{\text{H}_2}^{\text{ref}}}{1 + K_H x_{\text{H}_2}} \left(1 - \frac{Q}{K_{\text{st}}(T)}\right)_+, \quad (2)$$

in which c is the char concentration, $a(\varepsilon_o) = \varepsilon_o/(\varepsilon_o + 0.02)$ the fraction of pore surface reachable through open porosity ε_o , f_{cl} the closed-pore fraction, K_H an inhibition constant for hydrogen adsorption, $Q = x_{\text{CO}}x_{\text{H}_2}/x_{\text{H}_2\text{O}}$ the reaction quotient and K_{st} the equilibrium constant from NIST-JANAF data [47]. Char in closed pores is not gasified: once the pores seal, whatever carbon remains stays in the part. The reference atmosphere is 4% H_2 in nitrogen humidified at a dew point of 20°C , so that $x_{\text{ref}} = 0.023$ and $x_{\text{H}_2}^{\text{ref}} = 0.04$. The rate constant k_g is parameterised by a half-life $t_{1/2}$ at 800°C in that atmosphere and an activation energy; no measured value exists for char on either material, so both enter the sensitivity analysis. The square of the Thiele modulus, $k_g L^2/D_{\text{eff}} \sim 10^{-4}$ for $L = 2 \text{ mm}$, shows that gasification is kinetically controlled, so steam access is represented by the open-pore accessibility $a(\varepsilon_o)$ rather than by a diffusion equation.

Pyrolysis gas raises the pore pressure. The overpressure is computed from the compressible Darcy equation with Kozeny–Carman and Knudsen contributions to the permeability and a polymer-permeation term that carries the flux before the pore network percolates; it is compared with the green or brown strength through a safety factor. The same slab tracks the through-thickness temperature difference and converts it into a thermal-stress index.

2.3. Densification

Both materials densify according to the Skorohod–Olevsky viscous sintering model [48]. For a porous body of porosity θ

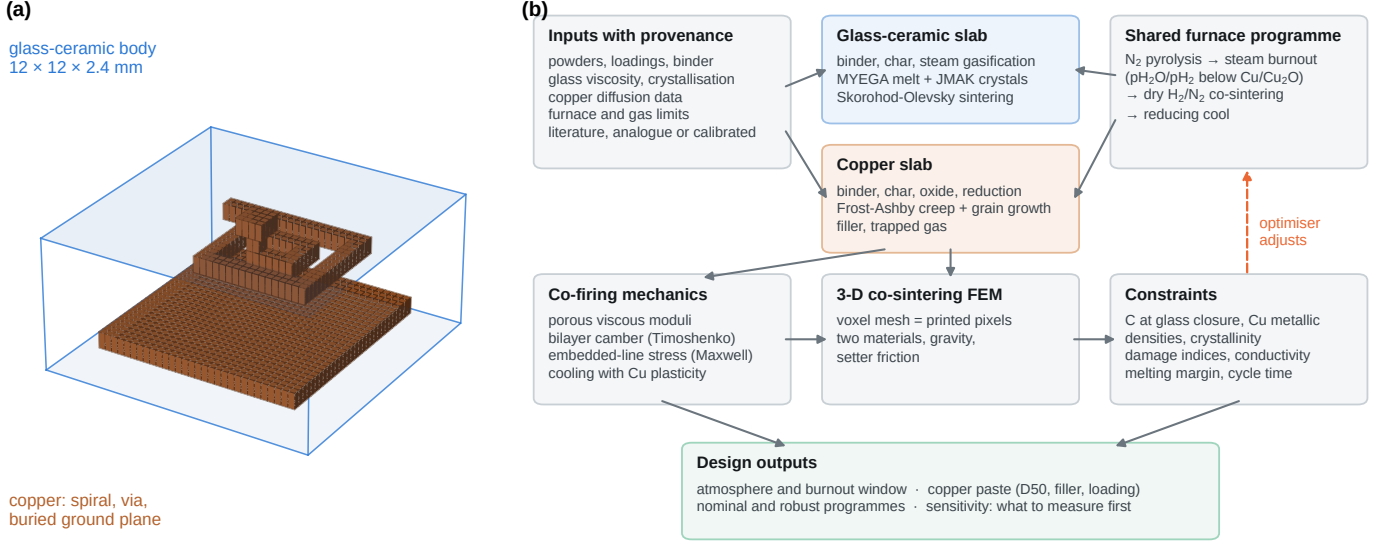


Figure 1: (a) Test vehicle: a 12 × 12 × 2.4 mm glass-ceramic body with a buried copper ground plane, a square spiral and a via to a top pad, printed as one part. (b) Structure of the model. Each material is followed by its own slab model under a shared furnace programme; the two histories are coupled mechanically, analytically and in three dimensions, and the programme and paste are searched against constraints.

and matrix shear viscosity η , the stress is

$$\boldsymbol{\sigma} = 2\eta [\varphi(\theta) \dot{\boldsymbol{\epsilon}}' + \psi(\theta) \text{tr}(\dot{\boldsymbol{\epsilon}}) \mathbf{I}] + P_L(\theta) \mathbf{I}, \quad (3)$$

with $\varphi = (1 - \theta)^2$, $\psi = \frac{2}{3}(1 - \theta)^3/\theta$ and sintering stress $P_L = 3\gamma(1 - \theta)^2/r$, where γ is the surface energy and r half the Sauter mean particle diameter. A free body densifies at $\text{tr}(\dot{\boldsymbol{\epsilon}}) = -(P_L - \Delta p)/(2\eta\psi)$, where Δp is the pressure of insoluble gas trapped in closed pores. The two materials differ only in η , γ and r .

Glass-ceramic. The residual melt is Newtonian. Its viscosity follows the MYEGA equation [49],

$$\log_{10} \eta_g = \log_{10} \eta_\infty + (12 - \log_{10} \eta_\infty) \frac{T_g}{T} \times \exp \left[\left(\frac{m}{12 - \log_{10} \eta_\infty} - 1 \right) \left(\frac{T_g}{T} - 1 \right) \right], \quad (4)$$

with $\log_{10} \eta_\infty = -3$ (Pa s), glass transition T_g (defined at 10^{12} Pa s) and fragility index m . Crystallisation during firing follows non-isothermal Johnson–Mehl–Avrami–Kolmogorov kinetics [50] written in terms of the extended-volume integral,

$$\dot{\xi} = k_c(T) = k_{c,p} \exp \left[-\frac{E_c}{R} \left(\frac{1}{T} - \frac{1}{T_{p,c}} \right) \right], \quad X = 1 - \exp(-\xi^n), \quad (5)$$

which avoids the singularity of the rate form at $X = 0$ for $n > 1$. The constant $k_{c,p}$ is fixed numerically so that the simulated crystallisation exotherm at 10 K min⁻¹ peaks exactly at the measured temperature $T_{p,c}$; Kissinger’s closed form [51] is exact only for $n = 1$ and would misplace the peak by about 1 K for $n = 2$. Crystals stiffen the melt as rigid inclusions,

$$\eta = \eta_g(T) \left(1 - \frac{X}{X_m} \right)^{-[\eta]X_m} \left(1 - \frac{f_c}{0.64} \right)^{-[\eta]0.64}, \quad (6)$$

the Krieger–Dougherty law [52] with intrinsic viscosity $[\eta] = 2.5$ and maximum packing X_m . The second factor treats residual char particles, volume fraction f_c , the same way. Once X approaches X_m the body stops flowing, which is how crystallisation arrests densification.

Copper. The copper skeleton creeps by diffusion. Its matrix viscosity combines lattice and grain-boundary diffusion [53],

$$\eta_{Cu} = f_\eta \frac{k_B T G^2}{42 \Omega (D_v + \pi \delta D_b / G)} \left(1 + \frac{C^2}{C_{inh}^2} \right) \left(1 - \frac{f}{0.64} \right)^{-1.6}, \quad (7)$$

where G is the grain size, Ω the atomic volume and f_η a calibration factor fitted to published copper densification data (Appendix A). The carbon term C (in ppm) describes a retarding effect of residual carbon on the sintering of copper. Its scale, $C_{inh} = 200$ ppm, is an estimate, and its influence on the results is examined in Section 4.10; and the last factor accounts for an inert ceramic filler of volume fraction f added to the conductor paste, a standard way to slow a metal paste down to the pace of its ceramic [39]. Grain growth is parabolic, driven by grain-boundary diffusion and pinned by porosity. The filler also lowers the conductivity; together with porosity,

$$\frac{\sigma_e}{\sigma_{e,0}} = (1 - \theta)^{3/2} \frac{1 - f}{1 + f/2}, \quad (8)$$

the second factor being the Maxwell Garnett result for insulating inclusions [54].

2.4. Atmosphere and the metallic state of copper

Steam removes char but oxidises copper once the water-to-hydrogen ratio exceeds the equilibrium of $\text{Cu}_2\text{O} + \text{H}_2 \rightleftharpoons 2\text{Cu} + \text{H}_2\text{O}$,

$$\frac{p_{\text{H}_2\text{O}}}{p_{\text{H}_2}} < K_{\text{Cu}}(T) = \exp \left(-\frac{\Delta G^\circ(T)}{RT} \right). \quad (9)$$

During the burnout segment the inlet hydrogen is therefore tied to the steam content, $x_{H_2} = x_{H_2O}/(s K_{Cu}(T_B))$, with a safety factor $s = 0.2$. The copper stays metallic by construction, and the steam is as oxidising towards carbon as the metal allows. This is the principle of the steam co-sintering developed for glass-ceramic/copper substrates [14, 6]; here it is applied in a model that says how long, at what temperature and in how much steam. Carbon gasification itself is thermodynamically unconstrained over the whole window (Section 4.2); its limits are kinetic.

2.5. Mechanics of co-firing

Porous viscosities. Eq. (3) describes an isotropic, linear viscous solid with shear viscosity $G_v = \eta\varphi$ and bulk viscosity $K_v = 2\eta\psi$, hence uniaxial viscosity $E_v = 9K_v G_v / (3K_v + G_v)$, viscous Poisson ratio $\nu_v = (3K_v - 2G_v) / [2(3K_v + G_v)]$ and biaxial viscosity $B = E_v / (1 - \nu_v)$. The free linear sintering rate is $\dot{\epsilon}_f = -P_L / (6\eta\psi)$. The viscosities and the free rate follow from the slab histories of T , θ , G or ξ , and carbon. Shrinkage is reported as engineering linear shrinkage when single materials are compared, and as logarithmic strain in the co-firing histories, where the free-strain mismatch is the difference of the logarithmic strains of the two materials.

Camber. By the elastic–viscous correspondence principle, a bi-layer of two linear viscous layers bends at a curvature rate given by Timoshenko’s bimetal solution [55], with elastic moduli replaced by biaxial viscosities and thermal strain by the difference in free sintering rates [56, 38, 41]:

$$\dot{\kappa} = \frac{6(1 + \lambda)^2 (\dot{\epsilon}_{f,2} - \dot{\epsilon}_{f,1})}{h[3(1 + \lambda)^2 + (1 + \lambda\mu)(\lambda^2 + 1/(\lambda\mu))]}, \quad (10)$$

with $\lambda = h_1/h_2$, $\zeta = B_1/B_2$ and $h = h_1 + h_2$. Layer 1 is copper and layer 2 the glass-ceramic; positive curvature is concave towards the copper. Eq. (10) is integrated over the firing with the layer thicknesses updated by their own shrinkage. The reference stack is a 0.1 mm copper plane on 1 mm of glass-ceramic. Curvatures above about 100 m^{-1} lie outside the small-deflection range of the solution and are used only to compare designs.

Embedded conductors. A conductor line buried in the body cannot curl; its misfit is taken up by stress. We represent line and surrounding body as two Maxwell elements in series, each with a porous elastic modulus and a biaxial viscosity:

$$\dot{\epsilon}_{f,GC} - \dot{\epsilon}_{f,Cu} = \dot{\sigma} \left(\frac{1}{E_{Cu}} + \frac{1}{E_{GC}} \right) + \sigma \left(\frac{1}{B_{Cu}} + \frac{1}{B_{GC}} \right). \quad (11)$$

A compliant line follows the body and carries B_{Cu} times the misfit rate; a stiff line forces the body to flow around it. Tension in the copper ($\sigma > 0$) opens voids and breaks the line; compression in the copper means the body is shrinking onto a stiff inclusion and is itself in hoop tension, the classic origin of radial cracks around conductors [36, 37]. Two damage indices compare these stresses with the strength of the partner that fails at that moment,

$$\Pi_{Cu} = \frac{\max(\sigma, 0)}{\sigma_{s,Cu}(\rho_{Cu})}, \quad \Pi_{GC} = \frac{\max(-\sigma, 0)}{\sigma_{s,GC}(\rho_{GC})}, \quad (12)$$

where the strengths σ_s grow from the binder-carried green strength through the neck-controlled brown strength. The series coupling carries an Eshelby factor of order unity [57] that we do not resolve, so the indices are used comparatively. The line damage index reported below is the larger of the two, $\Pi_{line} = \max(\Pi_{Cu}, \Pi_{GC})$. The three-dimensional model below resolves the same coupling in full.

Three-dimensional co-sintering. The printed part is meshed voxel by voxel with trilinear hexahedra, so the mesh follows the layout of the printed part at the resolution of the voxel. Each element carries its material, porosity and microstructural state. The velocity field satisfies $\nabla \cdot \sigma + \rho_b \mathbf{g} = \mathbf{0}$ with Eq. (3), gravity, and a setter plate beneath the part: zero normal velocity and regularised Coulomb friction ($\mu_f = 0.6$) iterated to convergence at each step. Volumetric terms use one-point quadrature, which avoids locking as $\theta \rightarrow 0$. The problem is assembled with scikit-fem [58] and solved by conjugate gradients preconditioned with smoothed-aggregation algebraic multigrid carrying the six rigid-body modes [59]. Porosity is advanced with the exact mass-conserving update $1 - \theta^{n+1} = (1 - \theta^n) \exp(-\text{tr } \dot{\epsilon} \Delta t)$. The temperature and carbon histories are taken from the two slab simulations. Below 0.3% porosity the constitutive law is evaluated at 0.3%, since the bulk viscosity otherwise grows without bound and the remaining densification no longer affects the shape. The simulation ends when the glass-ceramic reaches 98% density or 90% crystallinity, whichever comes first; the remaining linear shrinkage of the body is then a few tenths of a per cent.

Cooling. Below a stress-free temperature T_{sf} , where the residual glass stops relaxing, the thermal-expansion mismatch is carried elastically by the glass-ceramic and elastoplastically by the copper. The copper layer is treated as elastic–perfectly plastic with a temperature-dependent yield stress; its mean stress is capped at yield by a uniform plastic strain, and the curvature follows from Timoshenko’s elastic solution for the remaining elastic misfit.

2.6. Numerical solution, design search and sensitivity

The slab equations are integrated with a linearly implicit Rosenbrock method of orders 2(3) [60] on a geometrically graded mesh that is finest at the surface. A co-firing evaluation, two slab simulations and the mechanical post-processing, takes about 6 s. Designs were searched by differential evolution [61] over eleven variables: copper median particle size, filler fraction and solids loading; burnout temperature, time and steam fraction; temperature and duration of an intermediate matching hold; heating rate to the peak; peak temperature and hold. The objective was the largest free-strain mismatch between the two materials during firing, $\max_t |\epsilon_{f,GC} - \epsilon_{f,Cu}|$, which governs both camber and conductor stress. It was minimised subject to hard constraints and to a cycle-time budget of 18 h, the ϵ -constraint form of the trade-off between cycle time and mismatch. The constraints were: carbon at glass pore closure below 100 ppm and in the copper below 200 ppm; relative densities of at least 0.97 for the glass-ceramic and 0.92 for the copper; a crystallised

fraction of at least 0.8; a line damage index below 1; no trapped-gas pressure above 1.2 times the sintering stress; and a positive margin to the melting point of copper. Violations entered the objective as a penalty proportional to their size, so the search is steered back towards feasible designs. Sobol initialisation rounds the population up to a power of two, so 128 designs were evolved for 15 generations after initialisation (2048 evaluations). The objective still fell by 7% in the last generation, so the result is the best design found rather than a proven optimum; it is called the nominal optimum below.

A second, robust search asked for a design that works across the uncertainty of the two inputs that decide the timing, the gasification rate and the glass transition. It evaluates each design at two corners of that uncertainty. At the first, gasification is slow ($t_{1/2} = 4$ h) and the transition low ($T_g - 15$ K), so the glass tends to seal before the char has gone. At the second, gasification is fast ($t_{1/2} = 1$ h) and the transition high ($T_g + 15$ K), so the copper tends to be released before the glass is ready. The objective is the larger of the two peak mismatches; all constraints must hold at both corners, and the copper must in addition reach 80% IACS. The cycle-time budget was relaxed to 20 h, because the first feasible designs found needed about 20 h; the final design needs 16 h, so the budget did not bind. The search was seeded with one such feasible design, and a population of 128 was evolved for eight generations (1152 evaluations, each over both corners). Only the two corners described were used in the search; the other two corners were checked on the final design. The nominal case lies between the corners and was checked on the final design.

The influence of uncertain inputs was screened with Morris elementary effects [62]: 16 trajectories through a four-level grid over fourteen factors, 240 co-firing evaluations in total, around the nominal optimum. Twelve factors are the uncertain inputs with the ranges of Table 1; the other two are manufacturing tolerances on the copper paste, -25 to $+33\%$ in particle size and ± 0.05 in filler fraction. For each output the mean absolute elementary effect μ^* measures the overall influence of a factor and the standard deviation σ its non-linearity and interaction with other factors.

3. Parameters and calibration

3.1. Provenance of the inputs

No measurements were made for this study. Every input therefore comes from one of five sources, and Table 1 states which one applies. A value is labelled L when it was reported for the glass-ceramic of the IBM process or for copper itself; A when it was taken from an analogous material, usually a commercial copper-compatible LTCC; C when it was calibrated here against published process observations (Section 3.2); D when it follows from other entries of the table; and E when it is an engineering estimate. Most estimated and analogue values that matter enter the sensitivity analysis of Section 4.10, with the ranges given in the table; three that do not were checked one at a time.

The reference glass is the MgO-rich cordierite-type glass used by IBM for its copper-compatible substrates (glass #12: 52.5

SiO₂, 22.0 MgO, 22.0 Al₂O₃, 1.5 P₂O₅, 0.5 B₂O₃, 1.5 ZrO₂, wt%) [13, 63]. It was chosen because, among copper-compatible glass-ceramics, it is the one for which the firing process, the atmosphere and the reasons behind the chosen temperatures are best documented. We could not obtain published viscosity data for it, so its viscosity was calibrated.

3.2. Calibration of the glass against the IBM process

The patents that describe the IBM steam co-sintering process state, for glass #12, why the burnout hold sits where it does. The hold is placed at 785 ± 10 °C, between the annealing and softening points of the glass. At 750 °C carbon removal would take “prohibitively excessive amounts of time”; at 830 °C the glass pores close and trap water and binder residue [13]. A later IBM patent on the co-sintering of multilayer ceramic substrates states that 33–50% of the total fired shrinkage occurs during the burnout step [78]; we assume that this holds for glass #12, and a third places the onset of crystallisation at about 900 °C, after the burnout hold and before the crystallisation hold at about 960 °C [79]. We simulated the full worked schedule of Ref. [13] (N₂ to 200 °C, H₂/H₂O to 450 °C, 2.9 K min⁻¹ to the hold, 6 h at H₂/H₂O = 10⁻⁴, 0.5 h in N₂, 2.1 K min⁻¹ to 960 °C, 2 h) and used these statements as targets. The patent specifies the H₂/H₂O ratio but not the dilution; we assumed 50% steam in nitrogen.

Two of them fix the viscosity. Requiring that a 6 h hold at 780 °C produce 40% of the fired shrinkage with the pores still open, and that the pores close within 2 h at 830 °C, gives $\log_{10} \eta = 10.62$ at 780 °C and 9.37 at 830 °C. The MYEGA curve through these two points, with $\log_{10} \eta_\infty = -3$, has $T_g = 733.6$ °C and $m = 32.4$. The remaining observations were not used in the fit and serve as checks. Between 800 and 860 °C the calibrated viscosity falls by one decade per 43 K. Giess et al. found that the sintering of an IBM cordierite-type glass over the same range followed the viscosity of the glass, which fell by about one decade per 40 K [65]. The annealing point, taken as 10¹² Pa s, lies at 734 °C and the Littleton softening point (10^{6.6} Pa s) at 980 °C, so the 785 °C hold lies between them as the patent requires; so does the dilatometric softening point, near 10¹⁰ Pa s, which the calibrated glass reaches at about 803 °C. Stoichiometric cordierite glass is more viscous by 3.7 decades at 780 °C, 1.0 at 900 °C and 0.4 at 960 °C [80]. Glass #12 contains more MgO and less Al₂O₃ than cordierite, together with B₂O₃ and P₂O₅, so a lower viscosity is plausible, but this comparison is not a test of the fit.

The crystallisation kinetics were treated in the same way. With the activation energy measured on an IBM-type glass ($E_c = 303$ kJ mol⁻¹ [67]) and $n = 2$, placing the onset of crystallisation (5% crystallised) at 900 °C in the worked schedule requires an exotherm peak at 1040 °C at 10 K min⁻¹. The simulated exotherm then spans 978–1066 °C, a width of about 90 K, comparable to the 80–100 °C span of sintering temperatures that the IBM patent finds covered by the crystallisation exotherm of its glasses [63]. Differential thermal analysis of other MgO–Al₂O₃–SiO₂ glasses places the α -cordierite peak at 1006–1032 °C [64, 69].

Table 1: Model inputs. Basis: L, literature value for this material; A, analogue material; C, calibrated in Section 3.2; D, derived; E, estimate. Ranges in the last column are those used in the Morris screening.

Quantity	Symbol	Value	Basis	Source, remark or screening range
<i>Glass-ceramic body</i>				
Median particle size	d_{50}	3.0 μm	A	IBM glass powders 2–7 μm [63]; 2–7 μm
Size span $(d_{90} - d_{10})/d_{50}$	–	1.5	E	
Solids loading of the slurry	ϕ_{GC}	0.45	E	0.40–0.50
Density of the dense body	ρ_s	2600 kg m^{-3}	A	dense α -cordierite glass-ceramic, 2620 kg m^{-3} [64]
Surface energy	γ	0.36 J m^{-2}	L	calculated for a cordierite-type glass [65]; 0.24–0.36, since water vapour lowers the surface tension of silicate glass [66]
Glass transition (10^{12} Pa s)	T_g	734 $^{\circ}\text{C}$	C	714–754 $^{\circ}\text{C}$
Fragility index	m	32.4	C	28–40
High-temperature viscosity limit	$\log_{10} \eta_{\infty}$	–3	L	[49]
Crystallisation activation energy	E_c	303 kJ mol^{-1}	L	IBM-type high-cordierite glass [67]; 330–440 kJ mol^{-1} in other $\text{MgO-Al}_2\text{O}_3\text{-SiO}_2$ glasses [68, 69]; 250–470 kJ mol^{-1}
Avrami exponent	n	2	E	1.5–3.7; 2.8–3.7 in a related $\text{MgO-Al}_2\text{O}_3\text{-SiO}_2$ glass [69]
Exotherm peak at 10 K min^{-1}	$T_{\text{p,c}}$	1040 $^{\circ}\text{C}$	C	1000–1080 $^{\circ}\text{C}$
Maximum crystal packing	X_m	0.64	E	random close packing
Density at pore closure	ρ_{cl}	0.92	E	checked at 0.88 and 0.95
Thermal conductivity, dense	k	2.8 $\text{W m}^{-1} \text{K}^{-1}$	A	copper co-fired LTCC GL570 [70]
Young’s modulus, dense	E_{GC}	128 GPa	A	GL570 [70]
Poisson’s ratio	ν_{GC}	0.25	E	
Thermal expansion coefficient	α_{GC}	3.0 ppm K^{-1}	A	production glass-ceramic substrate [71]; 1.8 ppm K^{-1} for glass #12 fired in $\text{H}_2/\text{H}_2\text{O}$ [13]
Flexural strength	$\sigma_{\text{s,GC}}$	210 MPa	L	glass #12 fired in $\text{H}_2/\text{H}_2\text{O}$ [13]
Stress-free temperature	T_{sf}	690 $^{\circ}\text{C}$	D	strain point ($10^{13.5}$ Pa s) of the calibrated glass
<i>Copper conductor</i>				
Median particle size	d_{50}	1–20 μm	–	design variable; baseline 3 μm
Size span	–	1.2	E	
Solids loading	ϕ_{Cu}	0.40–0.60	–	design variable; baseline 0.50
Inert filler fraction	f	0–0.40	–	design variable; baseline 0
Viscosity factor	f_{η}	0.28	C	Appendix A, from [32]; 0.1–1.0
Lattice and boundary diffusion	$D_v, \delta D_b$	–	L	[53]
Carbon inhibition scale	C_{inh}	200 ppm	E	checked at 50 and 1000 ppm
Young’s modulus	E_{Cu}	117 GPa	L	oxygen-free copper [72]
Poisson’s ratio	ν_{Cu}	0.34	E	
Thermal expansion coefficient	$\alpha_{\text{Cu}}(T)$	16.5–23.7 ppm K^{-1}	L	instantaneous values, 293–1100 K [73]
Yield stress	$\sigma_y(T)$	69 MPa at 20 $^{\circ}\text{C}$	L	annealed temper [72]; softening $1 - T^{*1.09}$ [74, 75]
<i>Binder, char and atmosphere</i>				
Char yield of network and backbone	χ	0.05	E	0.02–0.10
Gasification half-life, 800 $^{\circ}\text{C}$, reference gas	$t_{1/2}$	2 h	E	0.5–8 h
Gasification activation energy	E_g	200 kJ mol^{-1}	A	order of steam gasification of carbons [76]; 160–240 kJ mol^{-1}
Hydrogen inhibition constant	K_H	25	E	form after [77]; checked at 5 and 100
Cu/Cu ₂ O and C/H ₂ O equilibria	$K_{\text{Cu}}, K_{\text{st}}$	–	L	NIST-JANAF [47]
Safety factor on the Cu/Cu ₂ O limit	s	0.2	E	
<i>Geometry</i>				
Slab half-thickness	–	2.0 / 0.15 mm	E	glass-ceramic / copper
Reference bilayer	$h_{\text{Cu}}, h_{\text{GC}}$	0.1, 1.0 mm	E	20 mm span

Table 2: The IBM worked schedule [13] simulated at three hold temperatures. Shrinkage is the fraction of the total fired shrinkage reached by the end of the 6 h hold; carbon is the residual carbon in the glass-ceramic at the end of the hold.

Hold temperature	750 $^{\circ}\text{C}$	785 $^{\circ}\text{C}$	830 $^{\circ}\text{C}$
Shrinkage during hold	7%	47%	100%
Pores at end of hold	open	open	closed
Carbon at end of hold (ppm)	33	0.006	0.07
Final relative density	0.990	0.990	0.990
Final crystallised fraction	1.00	1.00	1.00
Statement in [13]	too slow	chosen	pores close

Table 2 summarises the result. The 785 $^{\circ}\text{C}$ hold removes the char and leaves 47% of the shrinkage to the burnout step, inside

the 33–50% reported for the process, while the pores stay open. At 830 $^{\circ}\text{C}$ the body densifies completely during the hold and seals its pores. These two rows reproduce the calibration targets and are therefore not independent evidence. The 750 $^{\circ}\text{C}$ row is independent: it depends on the gasification kinetics, which were not fitted, and it leaves four orders of magnitude more carbon than the 785 $^{\circ}\text{C}$ hold. The model does not reproduce the absolute time scale of the patent’s statement. With the estimated gasification rate, 33 ppm of carbon remain after 6 h at 750 $^{\circ}\text{C}$, and whether that would have been acceptable cannot be judged, because no carbon specification was published. Nor does the model reproduce the reason the patent gives for avoiding 830 $^{\circ}\text{C}$: the carbon has gone before the pores close, and the trapped water that the patent mentions is not modelled. For all three holds the

final body is 99% dense and fully crystallised, and its free linear shrinkage is 23.1%.

4. Results

4.1. Verification

The model was checked against closed-form solutions before it was used for design (Appendix B). Isothermal densification of the glass-ceramic follows the Skorohod–Olevsky solution to within 1.3×10^{-4} in porosity for 4 to 16 cells (Fig. 2a), and the three-dimensional curvature rate of a free copper/glass-ceramic plate converges to the viscous Timoshenko value at least at second order, with an error of 0.05% at 0.15 mm voxels (Fig. 2b). The analytical camber of Eq. (10) and the finite-element model therefore describe the same mechanics, and the analytical form can be used inside the design search.

4.2. Atmosphere window for burnout

Fig. 3 places the burnout atmosphere in the plane of temperature and $p_{\text{H}_2\text{O}}/p_{\text{H}_2}$. Copper remains metallic below the orange line, which falls from $10^{6.2}$ at 500 °C to $10^{4.8}$ at 780 °C and $10^{4.1}$ at 1000 °C. Carbon gasification by steam is thermodynamically possible above the dashed line, which lies eight or more decades lower. The two constraints therefore leave a wide band of permissible atmospheres, and the choice within it is kinetic. At 780 °C, 30% steam with hydrogen at 20% of the copper limit ($p_{\text{H}_2\text{O}}/p_{\text{H}_2} = 1.2 \times 10^4$) gives a char half-life of 0.43 h. Forming gas (4% H_2) humidified through a bubbler at 20 °C, a common choice for copper, contains 2.3% water; its half-life at the same temperature is 3.1 h, seven times longer, because the steam partial pressure is low and hydrogen inhibits the reaction.

The operating point of the IBM process, $\text{H}_2/\text{H}_2\text{O} = 10^{-4}$ at 785 ± 10 °C [13], lies at 17% of the copper limit and almost coincides with the steam point in Fig. 3. The rule used here, hydrogen tied to steam so that the water-to-hydrogen ratio sits at a fixed fraction of the copper limit, is therefore consistent with the atmosphere used for tape-cast substrates. It also says how the ratio should change if the burnout temperature changes: by a factor of about 2.5 per 100 K, following $K_{\text{Cu}}(T)$.

4.3. Removing carbon before the glass seals

The char has to leave the glass before its pores close, and the glass carries far more of it than the copper does. After pyrolysis in nitrogen the glass-ceramic contains 22 500 ppm carbon relative to its solids, the copper 5 300 ppm. The difference follows from the densities: both slurries hold a similar volume of binder, but a unit volume of glass-ceramic solids weighs a third as much as the same volume of copper. Carbon on the glass is also the more harmful, since whatever is sealed in reacts with the water released during crystallisation and bloats the part [13].

Fig. 4 maps the carbon left in the glass-ceramic at the moment its pores close, as a function of burnout temperature and hold time. In steam (Fig. 4a) the 100 ppm contour runs from 16 h at 710 °C to 1.1 h at 830 °C, and the window closes abruptly at 850 °C, where the glass seals during the hold. The IBM hold of 6 h at 785 °C falls in the region with less than 10 ppm. In

humidified forming gas (Fig. 4b) only the longest hold, 16 h at 790 °C, brings the carbon down to 350 ppm; every other combination leaves more than 1000 ppm. Residual carbon also slows the densification of the glass. Acting as rigid inclusions, the 0.03–2% of carbon retained in forming gas lowers the final density from 0.99 to 0.96–0.99. For this material pair, steam is therefore necessary rather than merely helpful.

4.4. Two densification ranges

Fig. 5 compares the free sintering of the two materials after binder removal at 5 K min⁻¹ in dry 4% H_2 . The glass-ceramic shrinks by 23% between 850 and 933 °C (5 and 95% of its shrinkage) and then stops, because it crystallises: 93% of the glass has crystallised at 1060 °C. Fine copper densifies much earlier, and over a much wider range. A 3 µm powder shrinks between 454 and 702 °C; a 1 µm powder between 323 and 463 °C. Only for powders of 12 µm or coarser does the midpoint of densification fall inside the range of the glass, and these powders do not reach high density before 1060 °C: 83% for 12 µm and 60% for 20 µm.

The two curves also differ in shape. The glass densifies over a range of about 80 K, as a Newtonian melt whose viscosity falls steeply with temperature. Copper densifies over 140–350 K, because diffusional creep has a lower activation energy and grain growth slows it as it proceeds. Adding an inert filler to the paste delays and stretches the copper curve (Fig. 5b): 40% filler shifts the midpoint of 3 µm copper from 573 to 657 °C and widens the 5–95% range from 248 to 368 K. Neither particle size nor filler, on its own, gives copper a shrinkage curve that is as late and as steep as that of the glass. The matching has to come partly from the furnace programme.

4.5. Co-firing with an unmatched paste

The baseline design combines a 3 µm copper paste without filler with a programme that follows the IBM pattern: pyrolysis in nitrogen to 460 °C, a 4 h burnout at 780 °C in 30% steam, and sintering at 960 °C for 1 h in dry 4% H_2 (Fig. 6a). The design meets the chemical targets. The glass-ceramic retains 33 ppm of carbon when its pores close and reaches 99.1% density and full crystallinity; the copper is 97.6% dense, retains 81 ppm of carbon and conducts at 96% IACS.

It fails the mechanical ones. As soon as steam begins to remove the char, the copper starts to sinter, while the glass is still rigid. About an hour before the end of the burnout hold, the logarithmic shrinkage of the copper is 20.6% and that of the glass 4.9%. This is where the free-strain mismatch peaks, at 15.6%, and where a free bilayer of 0.1 mm copper on 1 mm glass-ceramic would reach a curvature of 58 m⁻¹, concave towards the copper. The damage index of an embedded line reaches $\Pi_{\text{Cu}} = 1.6$ at 680 °C, during the ramp to the burnout temperature, when the line is pulled in tension by the rigid body around it while its own necks are still weak. The glass then catches up, and its shrinkage overtakes that of the copper at 889 °C. The bilayer bends the other way and reaches -259 m⁻¹ at 930 °C. The final logarithmic shrinkages differ by 4%: 26.3% for the glass-ceramic against 22.3% for the copper. For this material

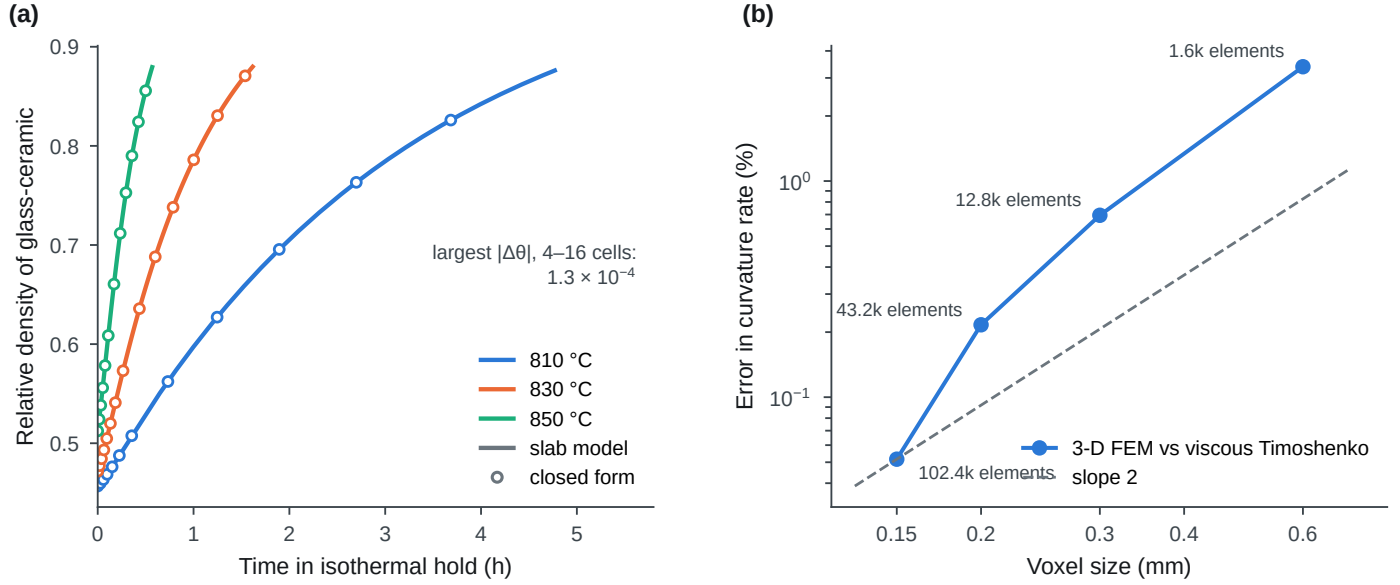


Figure 2: Verification. (a) Isothermal densification of the glass-ceramic slab at 810, 830 and 850 °C (lines, 8 cells) against the closed-form solution integrated along the simulated part temperature (circles); crystallisation and char were switched off. (b) Error of the three-dimensional curvature rate of a free 12 × 12 mm plate (0.6 mm copper on 1.8 mm glass-ceramic, 900 °C) against the viscous Timoshenko solution, Eq. (10), as the voxel size is refined.

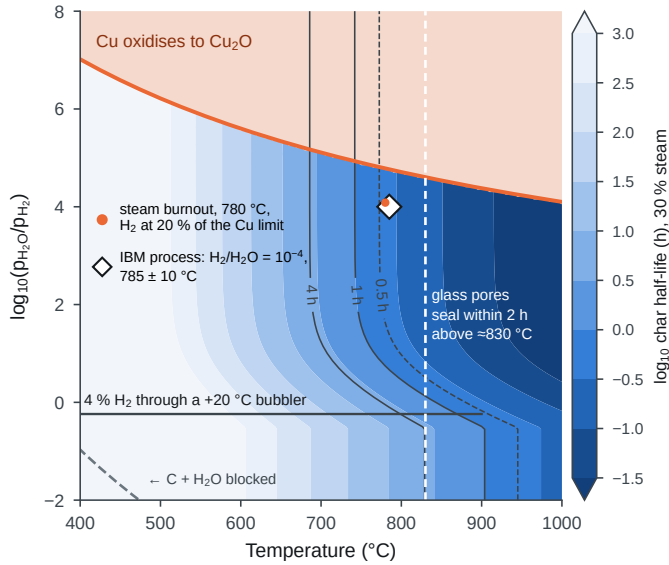


Figure 3: Burnout atmosphere window. Colour: half-life of char in 30% steam at the given p_{H_2O}/p_{H_2} (reference kinetics of Table 1). Copper oxidises above the orange line; carbon cannot be gasified to the left of the dashed line ($p_{CO} = 10^{-4}$ atm). Circle: steam burnout used in this work. Diamond: operating point of the IBM process [13]. The white line marks the temperature above which a 2 h hold seals the pores of the calibrated glass.

pair the difficulty lies less in the final shrinkage than in the order in which the two materials shrink.

4.6. Copper paste design maps

Fig. 7 shows how far the materials alone can repair the baseline, with the programme left unchanged. The upper row varies the copper paste. Coarser powder and filler both delay the copper, and the damage index of an embedded line falls below 1 for

particle sizes of 4.5 μm and above (Fig. 7b). The peak mismatch falls more slowly. Its lowest values, about 9.5%, are reached at 6 μm with 30% filler and at 9 μm without filler; coarser pastes are released too late and the mismatch rises again, now with the copper behind the glass. Every step towards a later copper costs density or conductivity (Fig. 7c,d). Filler lowers the conductivity directly, by 14 points of IACS for every 10 vol%, while coarse powder lowers it through porosity: 18 μm copper reaches only 76% density. Among the pastes that meet the density limit, 9 μm copper without filler gives the best compromise: 9.5% peak mismatch, a damage index of 0.38 and 91% IACS. None of them reduces the final camber, which stays between -190 and $-260 m^{-1}$ for every paste that meets the density limit. The final camber is set by the difference in final shrinkage, which the paste barely changes.

The lower row varies the glass, with the baseline paste. The glass transition decides whether carbon is trapped. Below about 695 °C the glass seals during the 780 °C burnout and retains about 3000 ppm (Fig. 7e). Above that, the carbon stays below 100 ppm, but the higher the transition, the later the glass densifies, and the crystallisation peak has to rise with it if the glass is to reach full density before it crystallises (Fig. 7f). The crystallisation peak must not be too high either: above about 1080 °C the glass is less than 80% crystalline after 1 h at 960 °C (Fig. 7g). The most interesting region is the narrow band around $T_g = 700$ °C (Fig. 7h). There the glass densifies during the burnout hold, together with the copper, and the peak mismatch falls to 6.5% while the carbon at closure stays near 50 ppm. A glass chosen to match the copper would thus do more than any paste, but only within a window of about 20 K in T_g , and with little margin on carbon at its lower edge.

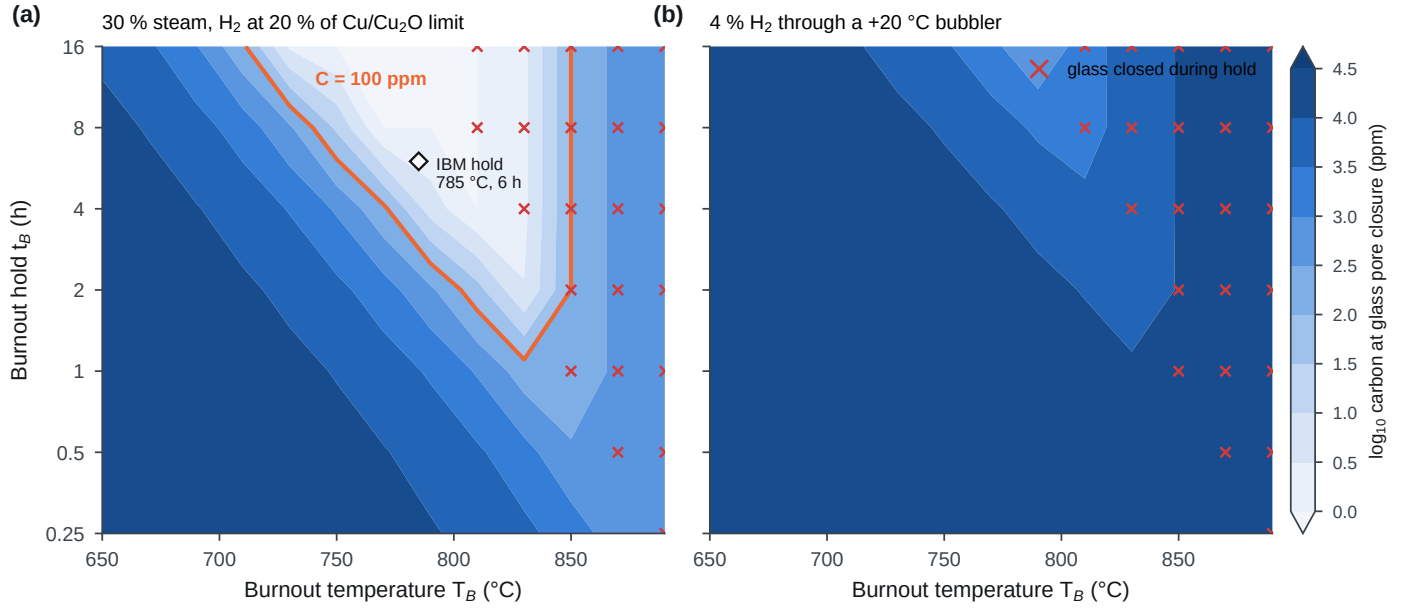


Figure 4: Carbon remaining in the glass-ceramic when its pores close, for burnout holds of temperature T_B and duration t_B within the baseline programme. (a) 30% steam with hydrogen at 20% of the Cu/Cu₂O limit; (b) 4% H₂ humidified at 20 °C. Orange: 100 ppm contour. Crosses: the glass sealed during the hold. Diamond: the IBM burnout hold [13].

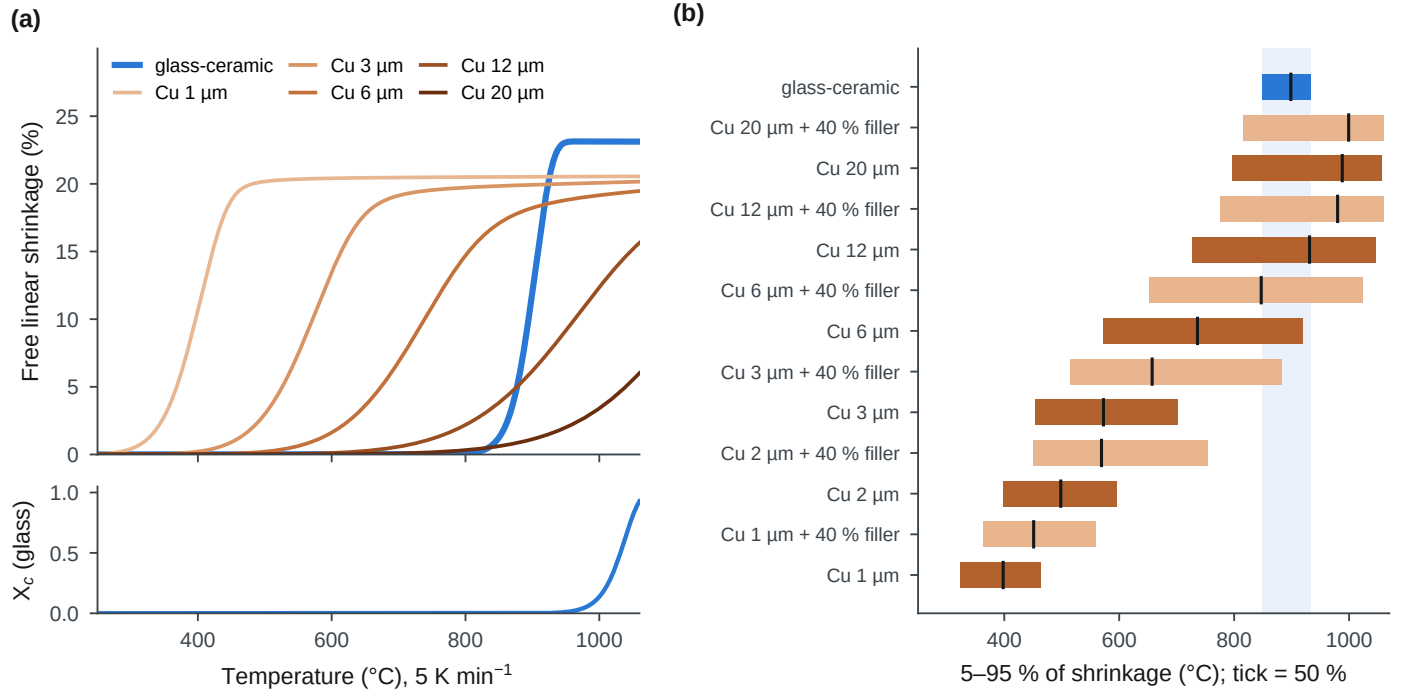


Figure 5: Free sintering of the two materials at 5 K min⁻¹ in dry 4% H₂ after binder removal. (a) Linear shrinkage of the glass-ceramic and of copper of different median particle size, and crystallised fraction of the glass. (b) Temperature range between 5 and 95% of the final shrinkage, with the midpoint marked, for copper with and without 40 vol% inert filler; the shaded band is the range of the glass-ceramic. Bars that reach 1060 °C had not finished densifying.

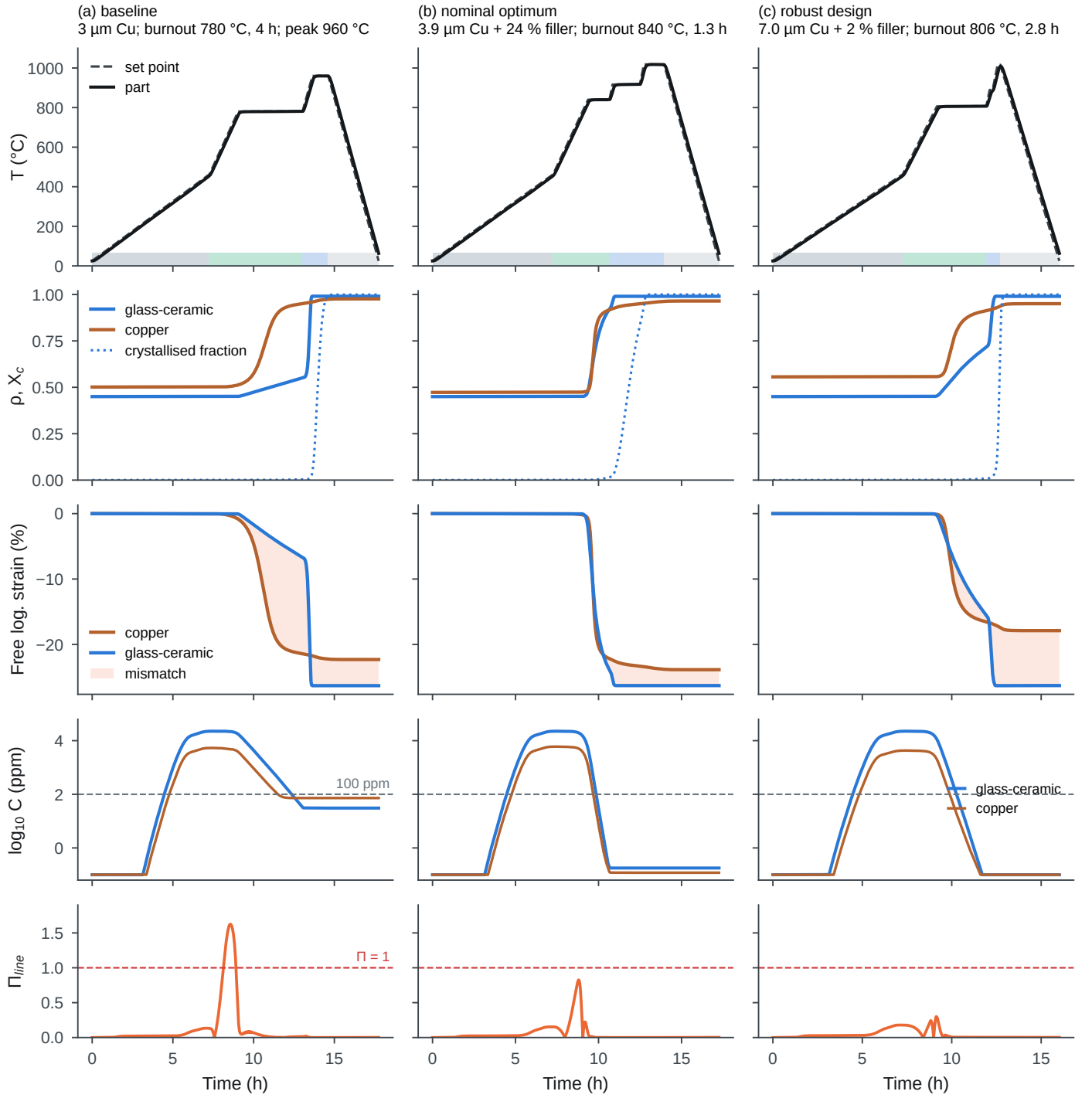


Figure 6: Co-firing histories of (a) the baseline, (b) the nominal optimum and (c) the robust design of Table 3. From top: furnace set point and part temperature, with the atmosphere of each segment shown by the band at the bottom (grey, N₂; green, steam; blue, dry H₂/N₂); relative densities and crystallised fraction of the glass; free logarithmic shrinkage of each material, the shaded area being the mismatch; carbon in the glass-ceramic and damage index of an embedded copper line.

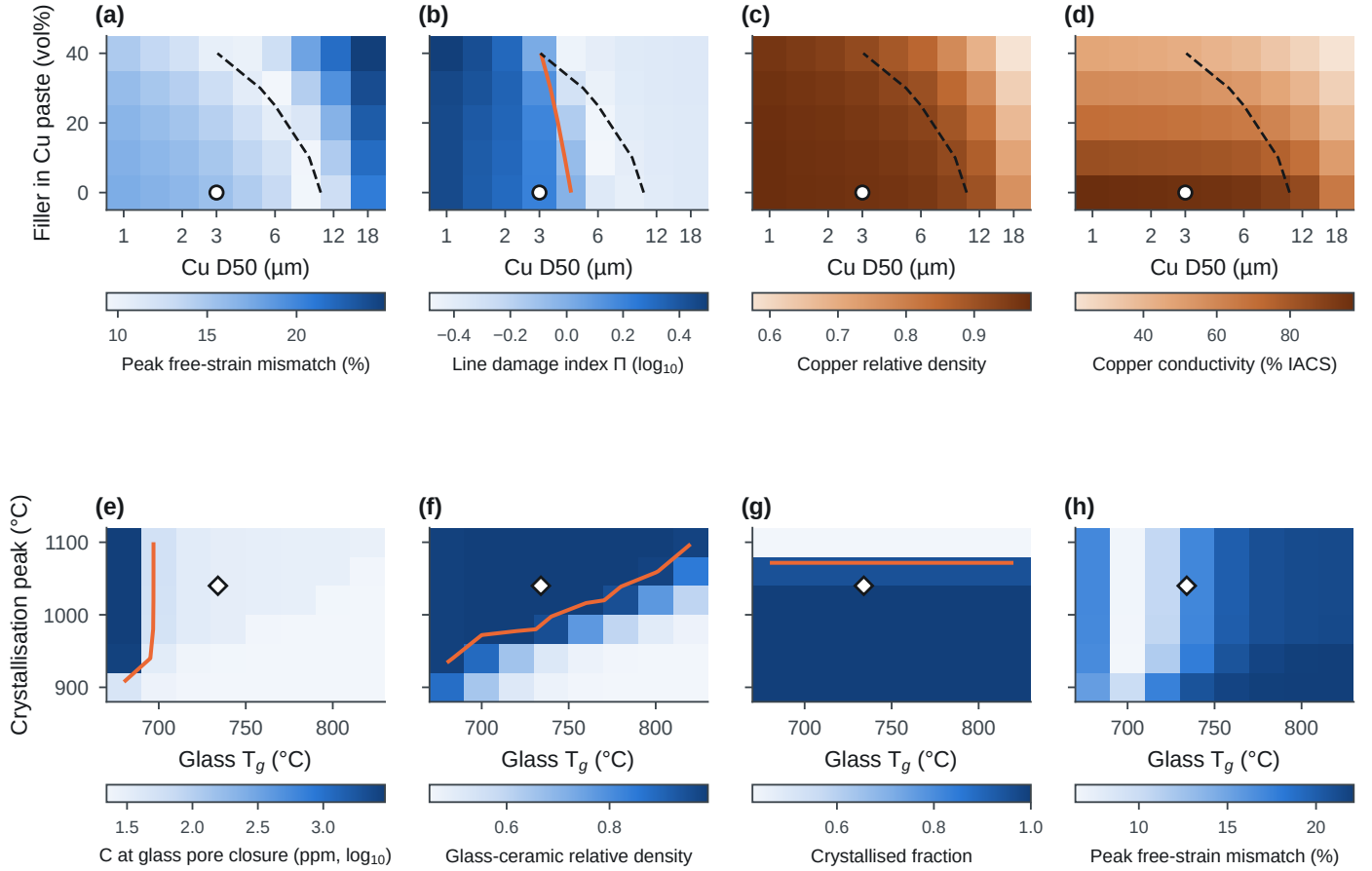


Figure 7: Design maps at the baseline programme. Top: copper paste, median particle size against filler fraction, with the baseline paste marked (circle): (a) peak free-strain mismatch; (b) damage index of an embedded line, orange contour $\Pi = 1$; (c) copper density; (d) conductivity. The dashed line is the density limit $\rho_{Cu} = 0.92$; pastes to its right fail it. Bottom: glass, transition temperature against crystallisation peak, with the calibrated glass marked (diamond) and the baseline paste: (e) carbon at pore closure, orange contour 100 ppm; (f) density of the glass-ceramic, orange contour 0.97; (g) crystallised fraction, orange contour 0.8; (h) peak mismatch.

4.7. A matched paste and programme

Neither the paste nor the glass alone brings the peak mismatch below about 6%, so the paste and the programme were searched together (Section 2.6). The best design found within an 18 h cycle, called the nominal optimum below, is listed in Table 3 and its history is shown in Fig. 6b. It uses a $3.9 \mu\text{m}$ copper powder with 24 vol% filler. The burnout is short and hot: 1.3 h at 840°C in 81% steam, just below the temperature at which the glass would seal during the hold (Fig. 4a). A 1.7 h hold at 916°C follows, and then 1.2 h at 1018°C . The peak mismatch falls from 15.6 to 3.7%, the damage index of an embedded line from 1.6 to 0.83, and the final camber of the reference bilayer from -254 to -76 m^{-1} . The glass-ceramic is 99.0% dense and fully crystallised, with 0.2 ppm of carbon at pore closure.

The nominal optimum works through the retarding effect of carbon on copper sintering that the model assumes (Eq. (7)). The copper cannot sinter while it carries char (Section 4.5), so the burnout releases it. Placing the burnout just below the coalescence of the glass, and making it fast with a high steam fraction, releases the copper at the moment the glass begins to flow. The two materials then densify together (Fig. 6b), and the filler slows the copper just enough to follow the glass rather than

lead it. The glass is already 99% dense when the hold at 916°C begins. The hold crystallises it, while the copper continues to densify, and the peak at 1018°C finishes the copper.

The optimum has two weaknesses. The first is conductivity: 24% filler limits the copper to 64% IACS. The second is robustness. The design is sensitive to the timing on which it rests, and the lower part of Table 3 shows what happens when uncertain inputs are varied one at a time. If the char gasifies half as fast as assumed, the carbon at closure rises from 0.2 to 76 ppm and the peak mismatch from 3.7 to 11.1%; at a quarter of the assumed rate the carbon limit is exceeded by an order of magnitude. A glass transition 15 K higher or lower than calibrated raises the peak mismatch to about 10%. So does any change in the carbon inhibition scale of copper sintering, C_{inh} , by a factor of four or five either way, and at 1000 ppm the damage index of a buried line exceeds 1. A factor of two in the viscosity of copper changes the mismatch less (5.8–6.2%), but halving it also takes the damage index above 1. A design that relies on the timing of char removal is therefore only as good as the gasification kinetics, the viscosity of the glass and the effect of carbon on copper, and none of them has been measured for this material pair.

The robust search (Section 2.6) therefore asked for a design

that meets every constraint at two corners of the uncertainty in gasification rate and glass transition and, in addition, keeps the copper above 80% IACS. The result, listed in the last column of Table 3 and shown in Fig. 6c, follows a different strategy. It uses a coarser 7 μm copper with almost no filler and a higher solids loading, and burns out for 2.8 h at 806 °C in 83% steam, some 40 K below the temperature at which the glass would seal during the hold. It then heats quickly to a short peak at 1017 °C and needs 16.0 h in all. Both materials start to densify during the burnout hold, the copper once its char has gone and the glass because it is already soft. Being coarse, the copper runs only a few per cent ahead of the glass; the glass overtakes it during the heating to the peak. At nominal inputs the peak mismatch is 9.3%, the conductivity 89% IACS and the damage index 0.30. In both search corners the carbon stays below 50 ppm, the damage index below 0.43 and the peak mismatch at or below 11.4%; the two other corners, not used in the search, give 9.1 and 9.7% with 34 ppm of carbon at most. Halving the gasification rate or shifting the glass transition by 15 K moves the peak mismatch only between 9.4 and 9.9%, and the design is equally insensitive to the carbon inhibition of copper and to the hydrogen inhibition of gasification (9.0–9.6%). Doubling the viscosity of copper is its weak point: the copper density falls to 0.912, just below the limit. The robustness is paid for in mismatch. The nominal value is two and a half times that of the nominal optimum, and the final camber, -187 m^{-1} , is larger. Most of the remaining mismatch is a difference in final shrinkage, 8.4%, because the paste now carries more solids than the glass slurry (56 against 45 vol%), which helps the copper reach density without filler. A gasification rate four times slower than assumed defeats both designs, which then leave 1295 and 940 ppm of carbon in the glass.

4.8. The printed test vehicle

The three designs were then fired in three dimensions as the printed test vehicle of Fig. 1a: a $12 \times 12 \times 2.4\text{ mm}$ glass-ceramic block with a buried $9 \times 9\text{ mm}$ copper ground plane, a 3.5-turn square spiral with 0.6 mm wide tracks, and a via to a 1.2 mm pad on the upper face. The block is meshed with 12 800 voxels of 0.3 mm and rests on a setter. The voxel is much coarser than the pixel of the printer, and every conductor is one voxel, 0.3 mm, thick: three times the copper of the reference bilayer, which exaggerates the distortion. Each run starts after binder removal and ends when the glass-ceramic is 98% dense, at 903–917 °C (Fig. 8). The setter keeps the lower face flat. The distortion therefore appears on the upper face, which dishes over the buried plane.

The buried plane appears to control the result, although no run without it was made. In the baseline the copper has already densified when the glass begins to flow, so the plane acts as a rigid inclusion. The block shrinks by only 14.9% in plane, against 23.1% for the free glass-ceramic, and the glass above the plane, unable to shrink laterally, densifies by shrinking more in thickness. The upper face ends 622 μm lower at the centre than at the rim (Fig. 8c), about a third of the fired thickness. In the nominal optimum the copper and the glass densify together. The plane then offers little resistance, the in-plane shrinkage rises

to 21.2%, and the relief of the upper face falls to 150 μm . The robust design lies between them: 18.3% in-plane shrinkage and 317 μm of relief. The relief follows the peak mismatch of the one-dimensional analysis rather than the final one: the robust design ends with a larger final mismatch than the baseline (8.4 against 4.0%) but with half its relief. The via and pad raise a small plateau at the centre of every part. That the three designs rank in the same order by relief and by peak mismatch supports the use of the one-dimensional analysis for design. The absolute values, however, depend on the area of copper. A plane that covers 56% of the part, as here, is the most demanding case, and a real layout would divide it or add dummy metal in the rim.

4.9. Cooling from the stress-free temperature

Below about 690 °C, the strain point of the calibrated glass, mismatch stresses no longer relax. The difference in expansion between copper ($16.5\text{--}22\text{ ppm K}^{-1}$ over this range) and the glass-ceramic (3 ppm K^{-1}) must then be carried by the solids. Fig. 9 shows the consequences for a copper plane on 1 mm of glass-ceramic. If copper remained elastic, its mean stress would exceed 900 MPa at room temperature and a 20 mm stack with 100 μm of copper would bow by 250 μm . Annealed copper cannot carry that stress. It yields within about 13 K of cooling and then follows its yield stress, which rises from about 27 MPa at 690 °C to 69 MPa at room temperature. The bow falls accordingly, to 13 μm for 100 μm of copper and to 6 μm for 50 μm , and the glass-ceramic stays in mild compression (-7 MPa for 100 μm of copper). The result does not depend on the stress-free temperature: starting the analysis at 500, 600, 700 or 800 °C gives the same final stress and bow, because the copper yields almost at once in every case. Plastic accommodation makes the thermal-expansion mismatch a secondary problem compared with the sintering mismatch, provided the copper is dense and ductile. The price is cyclic plasticity in the conductor, which matters for thermal cycling in service but is outside the scope of this study.

4.10. Which inputs decide the outcome

Fig. 10 ranks the fourteen uncertain inputs by their elementary effects around the nominal optimum. Elementary effects are expressed per full range of each factor. The half-life of char gasification dominates: across its range of 0.5–8 h it changes the peak mismatch by 16 percentage points on average and the carbon at glass closure by several orders of magnitude, from well below 1 ppm to above the 100 ppm limit. It also comes second for copper density, because char that is not removed keeps the copper from sintering. The carbon trapped in the glass depends next on the fragility and transition temperature of the glass, an order of magnitude less strongly. The density of the glass-ceramic is governed by its own properties: the transition temperature, the particle size and the crystallisation kinetics. The density and conductivity of the copper depend on the viscosity factor f_η and on the filler. The peak mismatch, the quantity that the design sought to minimise, responds to almost every input after the gasification rate: f_η , the glass transition, the char yield, and the particle size and solids loading of the glass. For every

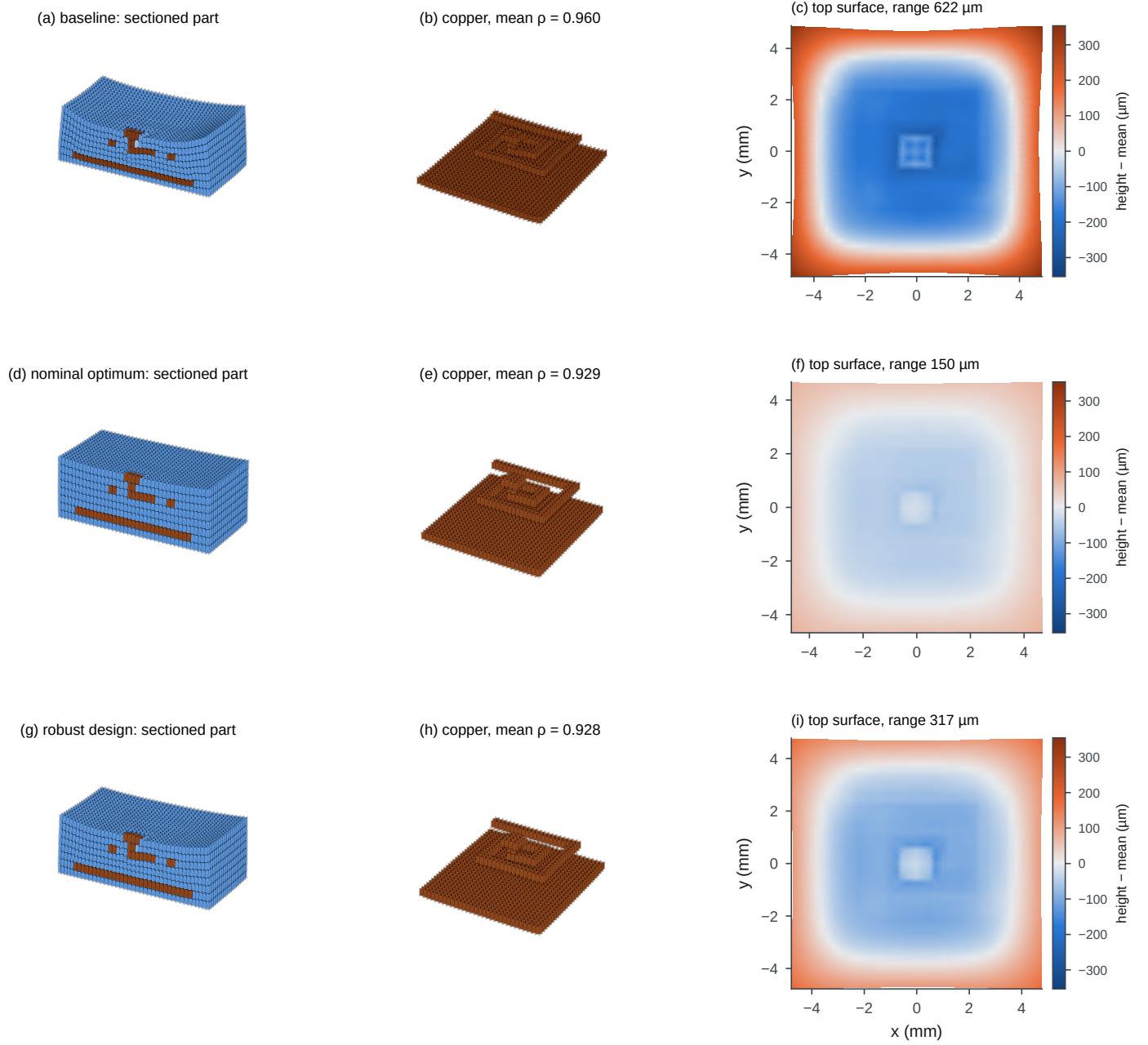


Figure 8: Three-dimensional co-sintering of the printed test vehicle (0.3 mm voxels, setter support, gravity) for the three designs of Table 3, from binder removal to 98% density of the glass-ceramic. Left: part sectioned through its centre, showing the buried plane, the spiral and the via. Centre: the copper, coloured by density. Right: height of the upper face relative to its mean; the lower face is held flat by the setter.

Table 3: Baseline and optimised designs. Lower part: the two optimised designs re-evaluated with one input changed from its nominal value: the char gasification half-life $t_{1/2}$ (2 h), the glass transition T_g (734 °C), the copper viscosity factor f_η (0.28), the carbon inhibition scale of copper sintering C_{inh} (200 ppm) and the hydrogen inhibition constant of gasification K_H (25). Bold: carbon above 100 ppm; [†]line damage index above 1; [‡]copper density below 0.92.

	Baseline	Nominal optimum	Robust design
<i>Paste and programme</i>			
Copper d_{50} (µm) / filler (vol%) / solids (vol%)	3.0 / 0 / 50	3.9 / 24 / 47	7.0 / 2 / 56
Burnout: T (°C) / time (h) / steam (%)	780 / 4.0 / 30	840 / 1.3 / 81	806 / 2.8 / 83
Matching hold: T (°C) / time (h)	–	916 / 1.7	889 / 0.18
Peak: T (°C) / time (h)	960 / 1.0	1018 / 1.2	1017 / 0.15
Cycle time (h)	17.7	17.3	16.0
<i>Outcome at nominal inputs</i>			
Carbon at glass pore closure (ppm)	33	0.2	0.03
Glass-ceramic density / crystallised fraction	0.991 / 1.00	0.990 / 1.00	0.990 / 1.00
Copper density / conductivity (% IACS)	0.976 / 96	0.965 / 64	0.950 / 89
Peak free-strain mismatch (%)	15.6	3.7	9.3
Line damage index max(Π_{Cu} , Π_{GC})	1.63	0.83	0.30
Final camber of reference bilayer (m ⁻¹)	–254	–76	–187
<i>Robustness: carbon at closure (ppm) / peak mismatch (%)</i>			
$t_{1/2} = 4$ h	–	76 / 11.1	37 / 9.9
$t_{1/2} = 8$ h	–	1295 / 17.9	940 / 14.4 [‡]
$T_g - 15$ K	–	84 / 9.9	0.06 / 9.4
$T_g + 15$ K	–	0.06 / 9.5	0.03 / 9.8
$f_\eta \times 2$	–	0.2 / 5.8	0.03 / 11.4 [‡]
$f_\eta / 2$	–	0.2 / 6.2 [†]	0.03 / 8.1
$C_{inh} = 50$ ppm	–	0.2 / 9.6	0.03 / 9.5
$C_{inh} = 1000$ ppm	–	0.2 / 9.8 [†]	0.03 / 9.1
$K_H = 5$	–	22 / 8.6	8 / 9.6
$K_H = 100$	–	0.0 / 10.2	0.0 / 9.0

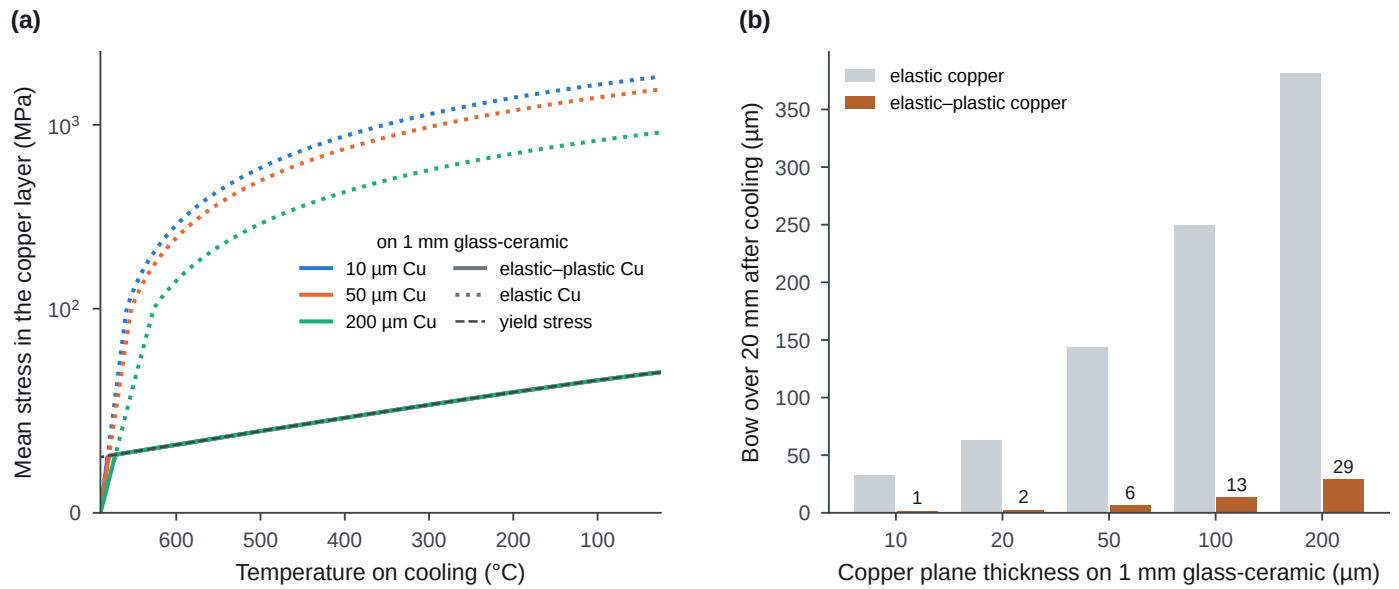


Figure 9: Cooling of a copper plane on 1 mm of glass-ceramic from the stress-free temperature (690 °C). (a) Mean stress in the copper layer for elastic and elastic–perfectly plastic copper; the plastic curves coincide with the temperature-dependent yield stress. (b) Bow over a 20 mm span at room temperature.

output that matters, σ is of the same order as μ^* . The effects are therefore strongly non-linear and interact, which is what a design based on the coincidence of two processes should be expected to show. The tolerances on the copper paste itself (particle size -25 to $+33\%$, filler ± 5 vol%) matter for the damage index of a buried line, together with f_h and the char yield, and the filler tolerance also for the conductivity. The Avrami exponent and the activation energy of crystallisation matter only for the density of the glass-ceramic, and the crystallised fraction (not shown) remained complete in every run. Three inputs were left out of the screening and checked one at a time instead (Table 3): the carbon inhibition scale of copper sintering, the hydrogen inhibition of gasification and the density at which the glass closes its pores. The last has little effect; the peak mismatch of the nominal optimum stays between 3.5 and 3.9%. The first two matter as much as the gasification rate for the nominal optimum, and hardly at all for the robust design.

5. Discussion

5.1. The order of events in co-firing

Taken together, the results fix an order of events that any firing programme for this material pair has to respect. Pyrolysis in nitrogen leaves about 2% of char, relative to the solids, in the glass-ceramic and about 0.5% in the copper. The char must then be gasified before the glass seals its pores. With the atmosphere held at a fixed fraction of the copper-oxide equilibrium, this requires hours at 770–800 °C and is the slowest step of the process. In the model, removing the char also releases the copper. Residual carbon is assumed to retard the sintering of copper (Eq. (7)), so a fine powder that has been held back by its own char starts to densify as soon as the steam clears it, at 680–780 °C in the baseline and well before the glass. This effect has not been measured for photopolymer char, and its strength, C_{inh} , is an estimate. In silver-metallised LTCC fired in air this coupling does not arise, because the binder burns off completely at low temperature and the metal is free to sinter from then on. Here the burnout step is also, whether intended or not, the first sintering step of the conductor. The glass then densifies over a range of about 80 K and stops when it crystallises, and below the strain point of the residual glass the expansion mismatch is taken up by plastic flow of the copper.

This order explains why neither lever works alone. The furnace programme cannot delay the densification of fine copper beyond the burnout, because the burnout has to happen below the coalescence of the glass. The paste cannot be made as late and as steep as the glass (Fig. 5) without losing conductor density. The two designs of Table 3 take different approaches. The nominal optimum uses the assumed coupling between char and copper to start both materials together. It reaches the lowest mismatch, but it depends on the timing of char removal and on the strength of the coupling, which are among the least known inputs of the model. The robust design gives up part of that gain and relies on a coarse copper that is intrinsically slow. It tolerates a factor of two in the gasification rate together with 15 K in the glass transition, and it is insensitive to the assumed effect of carbon

on copper. Which of the two is preferable depends on what has been measured. Once the gasification kinetics, the viscosity of the glass and the effect of carbon on copper are known much more tightly than these bounds, a design closer to the nominal optimum becomes safe.

5.2. Relation to tape-based co-firing

The rule used for the atmosphere, hydrogen tied to steam so that the water-to-hydrogen ratio sits at a fixed fraction of the copper limit, is consistent with the operating point of the IBM process (Fig. 3), and the calibrated glass is consistent with the burnout temperature chosen for it (Table 2). The model adds no new principle to that process, but it makes it quantitative: how long the burnout must last at a given temperature and steam content, how much carbon is left when the pores close, and how the choice of copper powder interacts with the programme. In LTCC, camber is controlled by adjusting the paste formulation [39] or, with master-sintering-curve predictions, the heating schedule [40]. For copper in a crystallising glass, Fig. 5 shows why paste formulation alone is not sufficient. The heating programme sets the order in which the materials are released to sinter, and the paste then adjusts the rate.

Printing changes the problem in three respects. First, a photocurable slurry carries more binder than a cast tape: at 45 vol% solids the glass-ceramic shrinks by 23% linearly, against 12.7–15.8% in the plane of commercial LTCC tapes [81, 82], and it holds correspondingly more char. Second, conductors are no longer confined to printed planes connected by punched vias. The test vehicle contains a spiral, a buried plane and a via defined in one build. In the three-dimensional model the large buried plane appears to control the distortion; tape technology would mesh or perforate such a plane, and a printed layout can divide it freely. Third, printed ceramics may shrink anisotropically between the build direction and the plane, as laminated tapes do: unconstrained IBM laminates fired in H_2/H_2O shrink about half as much through the thickness as in the plane [79]. The model treats both materials as isotropic. Anisotropy would change the camber and the shape of the fired part, but not the order of events described above.

5.3. Limitations

The results rest on published data and a model; no part has been fired. Four inputs deserve particular caution. The viscosity of the glass was calibrated against statements in patents, not measured, and it describes one glass composition; a different cordierite-type glass will shift the whole window. The gasification kinetics of photopolymer char on either material are unknown and were estimated, so the absolute times of the burnout carry the largest uncertainty of the study (Section 4.10). The retarding effect of carbon on copper sintering is an assumption of the model whose strength was estimated. The viscosity factor of copper was fitted to a coarser powder than those considered here. In addition, the model omits several phenomena that can matter in practice: dissolution of copper oxide in the glass and interdiffusion at the interface, adhesion and delamination, the water released by the glass during crystallisation, the

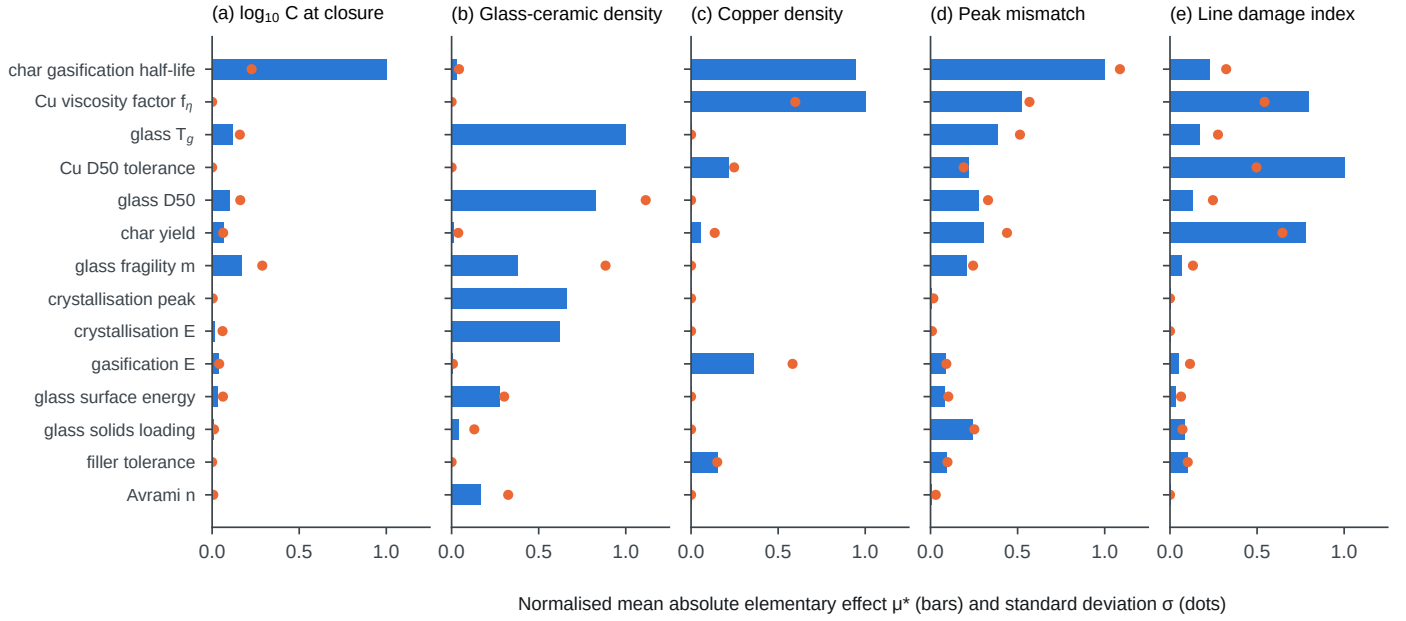


Figure 10: Morris screening of the fourteen uncertain inputs around the nominal optimum (16 trajectories, 240 co-firing evaluations). Bars: mean absolute elementary effect μ^* ; dots: standard deviation σ of the elementary effects, both normalised by the largest μ^* of each output. Factors are ordered by their summed normalised importance. Ranges as in Table 1; the copper particle size and filler fraction are varied as manufacturing tolerances around the design.

effect of residual carbon on dielectric loss, and cyclic plasticity of the copper in service. The damage indices compare stresses with strength estimates and are meant for ranking designs, not for predicting failure. The three-dimensional runs stop at 98% density of the glass-ceramic, so the final densification of the copper at the peak temperature, inside an already rigid body, is represented only in the slab model.

5.4. What to measure first

The sensitivity analysis (Fig. 10) and the one-at-a-time checks of Table 3 turn the limitations into a short list of measurements, ordered by their value to the design:

1. The gasification kinetics of the photopolymer char on each material, in the steam–hydrogen atmosphere of the burnout. Interrupted burnout runs followed by carbon analysis at three temperatures would fix both the rate and its activation energy. No other quantity has as large an effect on the carbon trapped in the glass and on the mismatch.
2. The viscosity of the glass between 750 and 900 °C, which fixes T_g and m . Beam bending or parallel-plate viscometry of the glass would give it directly, and so would isothermal dilatometry of the printed glass-ceramic, interpreted with the closed form of Appendix B.
3. The sintering viscosity of the copper slurry, f_η , from dilatometry of printed copper coupons in dry hydrogen after binder removal.
4. The char yield of both cured slurries, from thermogravimetry in nitrogen.
5. Whether, and how strongly, residual char slows the sintering of the copper slurry, from dilatometry of copper coupons after partial and complete burnout. The nominal optimum depends on it; the robust design does not.

The particle size and solids loading of the glass also matter, but they are specified by the supplier and the slurry formulation. The crystallisation kinetics matter little within the ranges considered, as long as the exotherm peak lies above the sintering range. All of these measurements are routine, and most of them can be made on single-material coupons printed with the same slurries.

5.5. Implications for three-dimensional packages

A fired body of this kind would have the properties that made glass-ceramic/copper attractive for high-end packaging. Its thermal expansion, 3.0 ppm K^{-1} , is close to that of silicon (2.6 ppm K^{-1}) [71], its permittivity is about 5 [13], and its conductors are copper. From porosity and filler alone the model predicts 89% IACS for the conductor of the robust design and 96% for a paste without filler. IBM reported $2.0 \pm 0.2 \text{ } \mu\Omega \text{ cm}$, about 86% IACS, for copper co-fired in $\text{H}_2/\text{H}_2\text{O}$ [13]. The difference points to impurities and grain-boundary oxygen, which the model does not include. Printing adds the geometric freedom that tape lamination lacks. Vertical coils, shielded lines, cavities and ground planes of arbitrary shape can be built into one part. That freedom is relevant to antenna-in-package modules and millimetre-wave front ends [3, 2], and to substrates for wide-bandgap power devices [4]. The present study defines a candidate firing route for such parts and the few quantities that have to be measured before it is tried.

6. Conclusions

A model of binder removal, sintering and crystallisation was used to design the co-firing of copper inside a cordierite-type glass-ceramic printed as one part by multi-material vat photopolymerisation. It was verified against closed-form solutions,

and its glass was calibrated against the published IBM glass-ceramic/copper process, with whose atmosphere and burnout temperature it is consistent. No experiments were made. The main conclusions are as follows.

1. Of the atmospheres examined, only steam removes the carbon from the glass before its pores close. Hydrogen tied to the steam at a fixed fraction of the Cu/Cu₂O equilibrium keeps the copper metallic and is consistent with the IBM operating point. At 780 °C such an atmosphere gasifies char seven times faster than humidified forming gas, which leaves more than 300 ppm of carbon in the glass-ceramic in every programme examined.
2. Fine copper and the glass densify in different temperature ranges: 450–700 °C for 3 µm copper against 850–930 °C for the glass. Co-fired with the IBM-type programme, the copper densifies during the burnout, the free-strain mismatch reaches 15.6%, and the damage index of a buried line exceeds 1. The final shrinkages differ by only 4%; the difficulty lies in the order in which the two materials shrink.
3. If char holds copper back, as the model assumes, the burnout step also releases the copper to sinter. The best paste and programme found exploit this coupling: a short, hot burnout just below the coalescence of the glass releases both materials together and reduces the peak mismatch to 3.7%. The design is sensitive to this timing. Halving the gasification rate, shifting the glass transition by 15 K or changing the strength of the carbon effect by a factor of four to five restores one half to two thirds of the mismatch that the optimisation removed, and the 24% filler limits the copper to 64% IACS.
4. A design required to meet every constraint under a factor of two either way in the gasification rate combined with ±15 K in the glass transition uses 7 µm copper with almost no filler and a 2.8 h burnout at 806 °C. At all four corners of that uncertainty it keeps the peak mismatch between 9.1 and 11.4%, the copper at 89% IACS and the carbon at glass closure below 50 ppm, in a 16 h cycle, and it is insensitive to the assumed effect of carbon on copper. The robustness costs a factor of two and a half in nominal mismatch.
5. In three dimensions, a buried copper plane that densifies before the glass acts as a rigid inclusion. The baseline test vehicle shrinks by 15% in plane instead of 23%, and its upper face dishes by 0.6 mm. The nominal optimum reduces the dishing to 0.15 mm and the robust design to 0.32 mm, the same ranking as the one-dimensional mismatch.
6. On cooling, plastic flow of the copper limits the stress it transmits to its yield stress. A 100 µm copper plane on 1 mm of glass-ceramic bows by 13 µm over 20 mm, about a nineteenth of the elastic prediction.
7. The inputs that decide the outcome are, in order, the gasification kinetics of the char, the viscosity of the glass, the sintering viscosity of the copper and the char yield. All four can be measured on single-material printed coupons, and they should be measured before the process is tried.

CRediT authorship contribution statement

[Author names]: Conceptualization, Methodology, Software, Formal analysis, Visualization, Writing – original draft, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The simulation code, input parameters with their provenance, and the scripts that generate every figure in this paper are available from the corresponding author on reasonable request.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the authors used [name of tool] in order to [reason, e.g. assist with code development, data analysis and drafting of the text]. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

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Appendix A. Copper slab model and its calibration

The copper conductor uses the slab model developed for single-material copper parts printed from a photocurable slurry. Besides the binder chemistry of Section 2.2, it tracks the oxygen bound in the powder as Cu_2O . The oxide grows in oxygen-bearing gas by a decelerating law fitted to powder oxidation data, is reduced by hydrogen above the equilibrium ratio of Eq. (9), and reacts with char by $\text{Cu}_2\text{O} + \text{C} \rightarrow 2\text{Cu} + \text{CO}$. None

of these oxidation paths is active in the co-firing programmes studied here, because the inlet gas contains no oxygen and the water-to-hydrogen ratio is held below the copper–oxide equilibrium. What matters for co-firing is the sintering viscosity. Eq. (7) uses the lattice and grain-boundary diffusion data for copper tabulated by Frost and Ashby [53], with one factor, f_η , fitted to the densification of copper parts printed from a 60 vol% photocurable slurry of 22 μm powder and sintered for 4 h at 1050 °C in hydrogen, which reached 90–94% relative density [32]. The fit gives $f_\eta = 0.28$, with 0.22–0.34 spanning the reported density range. The powder size enters Eq. (7) through the grain size, set initially to 0.4 times the median particle size, and through the sintering stress. Transferring f_η from 22 μm to the 3–12 μm powders considered here therefore rests on the diffusional scaling of Eq. (7); this is why f_η is included in the sensitivity analysis of Section 4.10.

Appendix B. Verification

Viscous densification of the glass-ceramic. With crystallisation and char removed, and open porosity, Eq. (3) with a Newtonian matrix gives $\dot{\theta} = -9\gamma\theta/(4r\eta)$. The slab model was run through isothermal holds at 810, 830 and 850 °C with 4, 8 and 16 cells. The simulated porosity follows the closed form, integrated along the simulated part temperature, to within 1.3×10^{-4} over the full range of densification before pore closure, independently of the mesh (Fig. 2a).

Bilayer curvature. The three-dimensional model of a free 12×12 mm plate made of 0.6 mm copper on 1.8 mm glass-ceramic was solved for the instantaneous velocity field at 900 °C, and the curvature rate was fitted to the mid-plane deflection. It converges to the viscous Timoshenko value of Eq. (10) at least at second order: the error falls from 3.4% at 0.6 mm voxels to 0.69, 0.22 and 0.05% at 0.3, 0.2 and 0.15 mm (Fig. 2b).

Free sintering and mass conservation. A free, frictionless cube held isothermally densifies homogeneously and isotropically, matches the closed-form Skorohod–Olevsky solution to 0.22% in density, and conserves solid mass to 0.07%.

Crystallisation kinetics. With the rate constant calibrated as described in Section 2.3, the simulated exotherm at 10 K min^{-1} peaks within 0.01 K of the prescribed temperature for Avrami exponents of 1, 2 and 3.